

Benjamin Antonio Correia

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EDUCATION

University of Massachusetts Amherst

Amherst, MA

BS in Informatics, Data Science Concentration | Minor in Computer Science (Major GPA: 3.7)

Graduation Date: December 2025

- Relevant Coursework: Data Science, Data Analytics, Statistics, Data Structures, Algorithms, Calculus, Database Management, Linear Algebra, Computer Systems, Business Intelligence, Statistics

EXPERIENCE

Business Insights & Analytics Leadership Development Program | Decision Science Intern

Hartford, CT

Travelers Insurance

June 2025 - Aug 2025

- Performed a detailed analysis of the influence of insurance quote turnaround time on conversion rate, utilizing data visualization and regression analysis to identify key trends and improve decision-making and task-queuing processes.
- Utilized Excel, SQL, R, Python, and other analytical tools to manipulate and analyze data, generating actionable insights.
- Supported business partners by providing reporting, dashboards, and analytical solutions to help them achieve their objectives.
- Communicated with business partners to clarify business goals, ensuring alignment and understanding to effectively tailor solutions that meet their needs.
- Assisted in the process of eliciting, analyzing, specifying, communicating, and validating business intelligence requirements while identifying project issues that require escalation.
- Engaged in comprehensive, industry-leading training focused on various technical subjects, leadership development, the insurance industry, and professional skills.

Machine Learning & Cybersecurity Intern

Waltham, MA

Dassault Systèmes | Solidworks

May 2024 - Aug 2024

- Researched and implemented machine learning techniques to detect network intrusions by monitoring network traffic.
- Transformed and refined large datasets using Pandas, Numpy, and Scikit-learn for improved model performance.
- Trained highly precise tree-based models for threat detection based on known attack signatures, achieving 99.9% accuracy with real-time detection capabilities.
- Developed a feed-forward neural network from scratch using Numpy and implemented an autoencoder in PyTorch to detect anomalous network traffic, potentially indicating new attack vectors.
- Implemented a Kalman filter algorithm from scratch to accurately estimate the position of location markers and improve the vision and pose estimation capabilities of a self-driving robot.

Data Science Undergraduate Course Assistant (TA)

Amherst, MA

University of Massachusetts Amherst

Sept 2024 - May 2025

- Assist with in-class activities and maintain awareness of course content by attending lectures.
- Grade class assignments, labs, and exams, ensuring consistent and accurate evaluation.
- Lead lab sessions and facilitate discussions to reinforce key concepts in statistics, data analytics, and machine learning.
- Monitor Piazza to provide timely assistance and answer student questions about coursework.
- Support coursework, which was entirely conducted in the R programming language.

PROJECTS

Analyzing Credit Risk Using Classification Models

- Utilized RMarkdown to generate a report on implementing logistic regression and random forest models to predict loan defaults.
- Assessed and compared the performance of multiple models using confusion matrix-based evaluation methods.
- Identified key predictors of loan defaults and employed subject matter expertise to explain the underlying reasons.

UMass Inauguration Simulator

- Developed an inauguration project simulation for the new chancellor of UMass Amherst, using multi-thread processing to allow "viewers" to call in and propose questions.
- Implemented a multithreaded approach with the creation of several threads using the pthread library in C
- Employed binary semaphores to control access to critical sections of code, including assigning caller IDs and checking for available call lines.
- Implemented semaphore destruction and thread termination to ensure proper cleanup and conclusion of the inauguration ceremony.

SKILLS

Skills/Competencies: Data Analytics | Predictive Analysis | Machine Learning | AI | Databases | Data Cleaning | Data Preprocessing | Data Science | Statistics | Business Intelligence | Algorithms | Data Structures | Data Visualization | Teaching | Insurance

Languages/Libraries: Python | Pandas | Scikit-Learn | Matplotlib | Numpy | PyTorch | SQL | R | RMarkdown | HTML | CSS

Tools: Microsoft Office | Excel | Powerpoint | VSCode | Jupyter Notebooks | Linux | Rstudio | SQLite Studio | Tableau | Git | Github | MicroStrategy | Qlik Sense | Snowflake